

CLOUD ENGINEERING BOOTCAMP • DAY 1

Computers, the Internet, and Cloud Computing

A foundations session for a complete beginner — how machines think, how they talk to each other, and why the cloud changed everything.

Duration
2–3 hours

Format
Guided walkthrough

Outcome
AWS-ready mindset

What today is building toward



Objective

By the end of Day 1, your friend should understand:

- How computers work
- How devices communicate over the Internet
- What cloud computing is
- Why companies use AWS instead of buying their own servers



Learning Outcomes — can your friend answer these?

What is a computer?

What is a server?

What is the Internet?

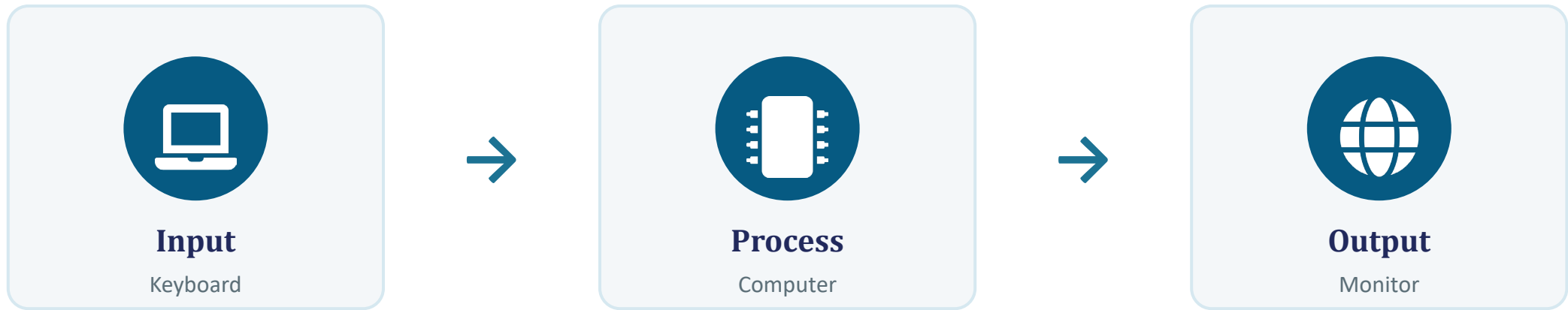
How does a website load?

What is cloud computing?

Why do companies use AWS?

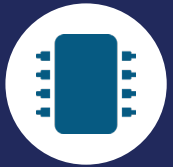
What is a computer?

A machine that receives input, processes data, stores data, and produces output.



Ask your friend: “What happens when you open Google Chrome?” — this is the activity for this section.

Main components of a computer



CPU

The “brain”

Executes instructions and performs calculations. Opening Chrome triggers thousands of CPU instructions.



RAM

Temporary memory

Think of RAM as your desk — more RAM means a bigger desk. Everything in it disappears on shutdown.



Storage

Permanent memory

SSD or HDD. Stores Windows, games, documents, and pictures for the long term.

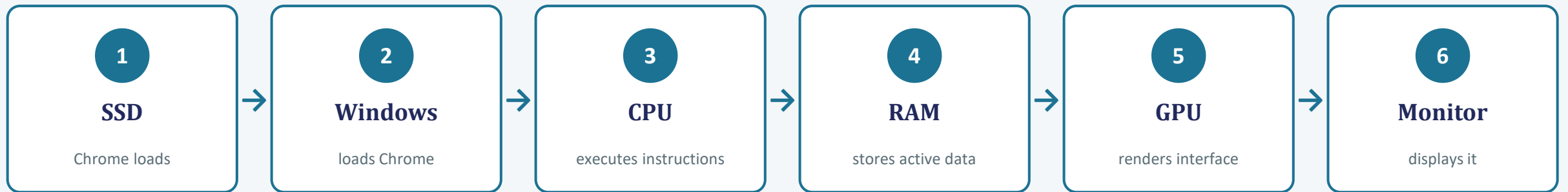


GPU

Graphics processor

Renders visuals — essential for gaming, AI workloads, and video editing.

What happens when you open Chrome?



Guide your friend toward this exact sequence — it's the mental model for everything that follows.

What is an Operating System?



The manager of the computer

An Operating System manages hardware and allows applications to run.

Without an OS, nothing works.

EXAMPLES

Windows

Linux

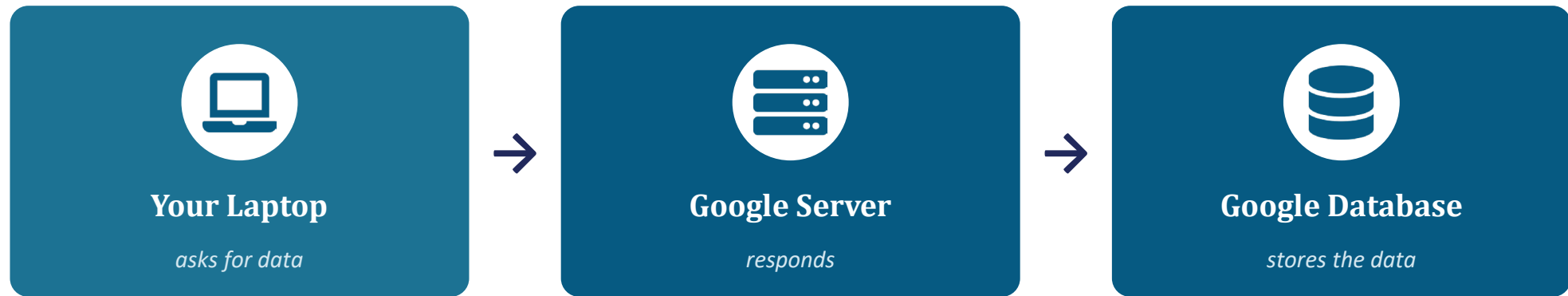
macOS



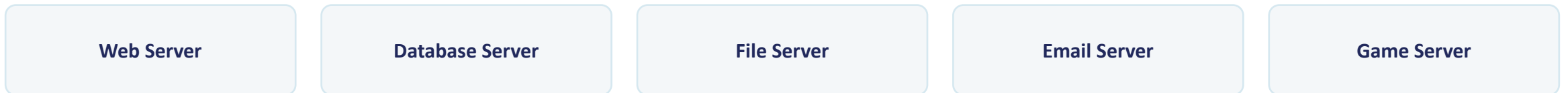
*“Think of it as the
manager of the computer.”*

Client vs. server

Most people only use clients. Cloud engineers manage servers.



EXAMPLES OF SERVERS



Ask: “Is Facebook running on your laptop?” No — it runs on thousands of servers around the world.

What is the Internet?

Millions of computers connected together.



Every website works this way.

WHEN YOU TYPE `www.google.com`



Vocabulary you'll use every day



Website

A collection of web pages.



Browser

Chrome, Firefox, Edge,
Safari.



ISP

Internet Service Provider —
e.g. PLDT, Globe, Converge.



DNS

The Internet's phonebook.
Converts google.com into
an IP address.



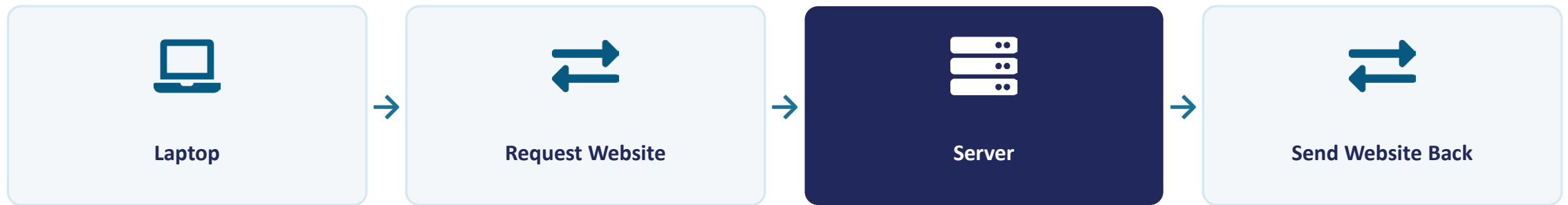
IP Address

Every device has one, e.g.
192.168.1.100.

Lab moment: run `ping google.com` and `nslookup google.com` in Command Prompt to see DNS resolve a name to an IP in real time.

What is a server?

A server is simply another computer — the difference is that it serves requests.



SERVERS USUALLY HAVE



What is cloud computing?

Imagine opening a coffee shop.

OPTION 1

Buy everything yourself

- Server
- Networking
- Cooling
- Electricity
- Security
- Maintenance

Cost: millions of pesos over time

OPTION 2

Rent from AWS

- Servers
- Networking
- Buildings
- Cooling
- Security

You simply rent what you need

Cloud computing means: using computing resources over the Internet without owning the physical infrastructure.

Why companies use AWS



Pay only for what you
use



Scale up or down
quickly



Deploy globally



High reliability



Strong security features

AWS IN EVERYDAY LIFE

Netflix

Streaming

Airbnb

Booking

Samsung

Some services

Adobe

Cloud services

Many large companies use multiple cloud providers, but AWS is one of the most widely adopted.

Suppose you create an online store

WITHOUT AWS

- 1 Buy Servers
- 2 Rent Building
- 3 Maintain Hardware
- 4 Replace Failed Components

WITH AWS

- 1 Create EC2
- 2 Deploy Website
- 3 Done

AWS manages the underlying hardware for you.

Mini quiz

1 What is the brain of the computer?

2 What is RAM used for?

3 What is the difference between RAM and SSD?

4 What is an Operating System?

5 What is a server?

6 What is the Internet?

7 What is DNS?

8 What is an IP address?

9 What is cloud computing?

10 Why do companies use AWS?

Open Command Prompt and run:

```
hostname
```

What is your computer's name?

```
ipconfig
```

Find your IPv4 Address, Default Gateway, and DNS Server.

```
ping google.com
```

Observe the IP address returned, packet loss, and round-trip time (latency).

```
nslookup google.com
```

Identify which DNS server answered and the IP address(es) returned.

Before Day 2

Watch videos on:

- How the Internet Works
- What is Cloud Computing?
- AWS Global Infrastructure (Regions and Availability Zones)

Draw from memory:



Reflect in your own words:

- Why doesn't your laptop store every website you visit?
- Why is DNS necessary?
- What advantages does AWS provide compared to buying your own servers?

Before moving to Day 2

Your friend should be able to explain, without notes:



The role of the CPU, RAM, storage, and operating system.



The difference between a client and a server.



How a website request travels from a browser to a server and back.



What DNS and IP addresses are.



What cloud computing is and why businesses use AWS.

This conceptual foundation makes the networking and Linux topics ahead much easier to understand.